Job Sequencing and Tool Switching

Structural results and a branch-and-Benders-cut study

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The order decides which tools return

Magazine capacity $b = 2$; the machine starts empty.

order $1, 2, 3$ $\quad J_1\ \{a, b\} \xrightarrow{+c} J_2\ \{b, c\} \xrightarrow{+d} J_3\ \{c, d\}$ $\quad$ 4 **insertions**

start: $+a, +b$

order $1, 3, 2$ $\quad J_1\ \{a, b\} \xrightarrow{+c,\, +d} J_3\ \{c, d\} \xrightarrow{+b} J_2\ \cancel{\{b, c\}}$ $\quad$ 5 **insertions**

start: $+a, +b$

b was removed, then returned

Report: Figures 1–2 and Section 1.1.

Loading is easy; the order is the hard part

Fixed order

When a tool must leave, remove the loaded tool whose next use is furthest away.

$$\mathrm{KTNS}(\pi) \quad \text{in} \quad O(n|U|).$$

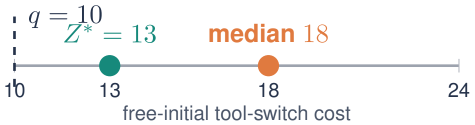
Choosing the order is NP-hard for every fixed $b \geq 2$.

$40{,}320$

orders of the eight-job instance `L1-1`

180

are optimal



Tang–Denardo KTNS; Crama et al. hardness; report Figure 3.

Every stop costs production time

CNC cells	**PCB assembly**	**Colour printing**	**Order picking**
tool magazine	feeder slots	ink cartridges	what a picker carries

Same combinatorial core

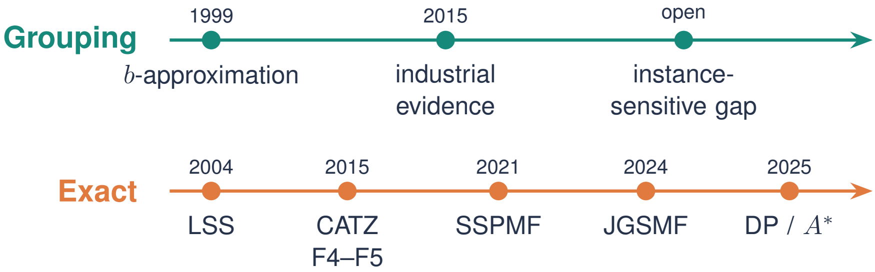
A capacity-b resource, jobs that need subsets of it, and a changeover cost paid between consecutive jobs.

Why the order matters

A cell running thousands of parts a day repeats the changeover constantly, so the sequence is chosen once and paid for every shift.

Report: Section 1.1; industrial study of Burger et al. (2015).

Two gaps the literature leaves open



Two things the timeline does not contain: a bound on the two-phase loss that depends on the **instance**, and a relaxation that gets **past the counting bound**.

Report: literature review in Chapter 3.

Two questions, and how the talk is organised

Question 1 – structural

Industry does not solve the SSP. It groups jobs into as few magazine-fitting batches as possible, then sequences the batches.

What does insisting on the fewest groups cost, once the final KTNS step is allowed to re-optimise the loading?

Question 2 – algorithmic

Every compact model surveyed relaxes to the counting bound q, so it is fast where q is exact and adrift where it is not.

Can a method that never forms that relaxation – an exact loading oracle, or a reformulation over configurations – get past it?

Part I answers the first and finds an unbounded loss. **Part II** answers the second, and the answer is no — for a reason worth naming.

Report: Section 1.3, questions set by the advisors.

The configuration view, and where q comes from

$$\{a,b,c\} \xrightarrow{+d} \{b,c,d\} \xrightarrow{+e} \{c,d,e\} \xrightarrow{+f} \{d,e,f\}$$

jobs served here   jobs served here

A schedule is a walk

Magazine configurations are the nodes; the step cost is $d(C, C') = |C' \setminus C|$, the tools that must be inserted. Every job must be served at some node that contains its requirement.

Hence one counting bound

Each used tool enters at least once, and at most b of them can already be loaded at the start:

$$Z^* \;\geq\; q \;=\; |U| - b.$$

This single quantity governs **both** halves of the talk.

Report: Section 2.4, configuration graph; Corollary 1.

Grouping commits before it sees the sequence

Jobs

$T_j \subseteq U$

→

Minimum grouping

$\left|\bigcup_{j \in G} T_j\right| \le b$

fewest groups $= K^*$

→

Group and job order

best choices over every tie

→

KTNS cost H

best final loading for that order

$$Z^*$$

best KTNS cost over every job order

$$H - Z^*$$

loss that remains even under favourable tie-breaking

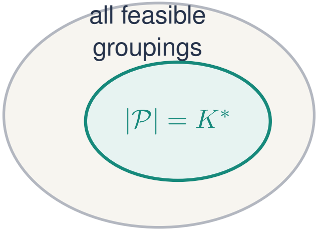
Report: Definitions 5–7.

Drop the fewest-groups rule: grouping is exact

$$Z^* = \min_{\mathcal{P}\ \text{feasible}} \gamma(\mathcal{P})$$

$$H_{\text{walk}} = \min_{\substack{\mathcal{P}\ \text{feasible} \\ |\mathcal{P}|=K^*}} \gamma(\mathcal{P}), \qquad H \le H_{\text{walk}}.$$

The approximation enters when the first phase keeps only the groupings with the fewest groups.



$$\text{cost} = q + R$$

$q = |U| - b$; R counts reinsertions.

Report: Theorem 1, Corollary 1 and Lemma 1.

Four jobs are enough for a genuine grouping loss

Required tools; capacity $b = 4$

	1	2	3	4	5	6	7
A	●	●					
B			●	●			
C	●		●		●	●	
D		●			●	●	●

A and B are the only feasible pair.

Unrestricted order

$$B, C, A, D \quad \Longrightarrow \quad Z^* = q = 3.$$

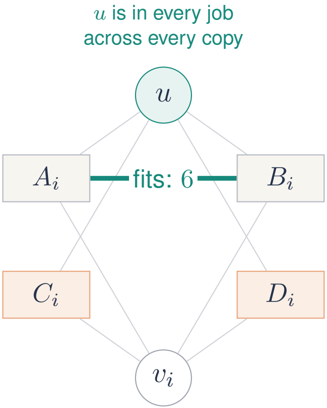
Every minimum grouping

$$A \text{ and } B \text{ consecutive} \quad \Longrightarrow \quad H = 4.$$

One forced adjacency creates one reinsertion.

Report: Proposition 10.

One shared tool makes the family connected



- C_i and D_i fill all six slots.
- $A_i \cup B_i$ uses exactly six tools.
- Small jobs from different copies use seven.

unique minimum grouping

$$K^* = 3g$$

Report: Theorem 5, construction and first proof step.

The unrestricted order reaches coverage

$$B_i \to C_i \to A_i \to D_i$$

u remains loaded

repeat copy by copy; every newly loaded private tool is appearing for the first time

$$|U| = 8g + 1, \qquad b = 6$$

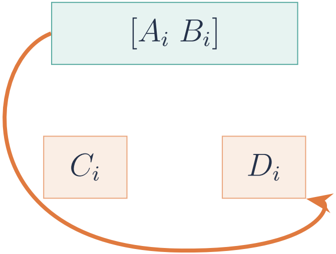
$$q = |U| - b = 8g - 5.$$

No reinsertion is needed, so the coverage lower bound is attained:

$$\boxed{Z^* = q = 8g - 5}.$$

Report: Theorem 5, coverage schedule.

Every minimum grouping forces one return per copy



at least one private tool must return

Lower bound

$$H \geq q + g = (8g + 1 - 6) + g = 9g - 5.$$

Matching grouped order

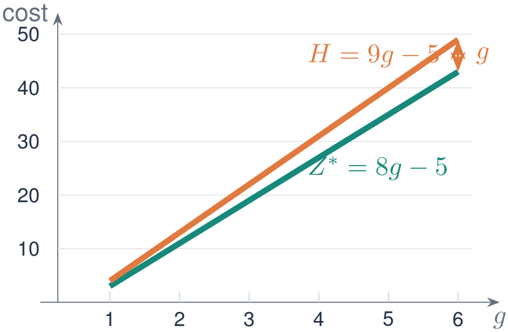
$$A_i, B_i, C_i, D_i$$

has exactly one return per copy.
Therefore

$$\boxed{H = 9g - 5}.$$

Report: Theorem 5, forced-return count and matching schedule.

The additive gap grows without bound



$$H - Z^* = g$$

The additive loss is unbounded, even though all jobs remain connected through u.

$$\frac{H}{Z^*} \longrightarrow \frac{9}{8}.$$

This family does not prove a universal $9/8$ ratio bound.

Report: connected unbounded heuristic-gap theorem.

Two routes past the counting bound

The obstacle
every compact model surveyed relaxes to q or below

Route 1: decompose
never relax the loading at all

Route 2: reformulate
optimise over configurations

Branch-and-Benders-cut

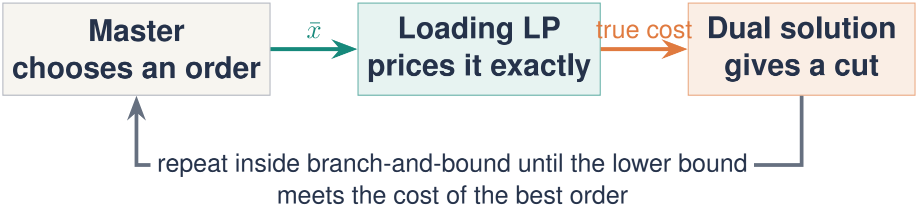
The loading subproblem is polynomial, so it becomes an exact oracle and the master never forms a weak relaxation *of the loading*.

Position-indexed branch-and-price

Columns are magazine configurations. One variant prices *transitions*, and its relaxation can provably exceed q.

Report: Section 5, opening; Table 3.

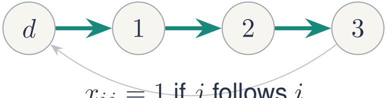
Order in the master, loading in an exact oracle



The decomposition is natural because the fixed-order subproblem is polynomial and exact.

Report: Figure 9 and Section 5.2.

The master starts from local and coverage bounds



$x_{ij} = 1$ if j follows i
d marks the start and end

$$\min\ \theta, \qquad \sum_{i \neq j} x_{ij} = 1, \qquad \sum_{j \neq i} x_{ij} = 1,$$

$$\sum_{i,j \in S} x_{ij} \leq |S| - 1 \qquad (S \subseteq J).$$

Starting lower bounds

$$\theta \geq |U|, \qquad \theta \geq \sum_{i,j \in J} w_{ij} x_{ij} + \sum_{j \in J} |T_j| x_{dj},$$

where $w_{ij} = \max(0, |T_i \cup T_j| - b)$.

Report: Benders master formulation, Section 5.2.

The loading LP is integral on a fixed path

$$
\begin{aligned}
\min \quad & \sum_{j,t} z_{jt} & \\
\text{s.t.} \quad & y_{jt} \ge 1 & (t \in T_j), \\
& \sum_{t} y_{jt} \le b & (j \in J), \\
& z_{jt} \ge y_{jt} - y_{it} - (1 - x_{ij}) & (i \ne j), \\
& z_{jt} \ge y_{jt} - (1 - x_{dj}) & (j \in J), \\
& y, z \ge 0. &
\end{aligned}
$$

y_{jt} records whether tool t is loaded at job j.

z_{jt} charges an insertion when the selected path moves into j.

fixed path

$\Downarrow$

exact KTNS cost

Report: fixed-order loading formulation and integrality argument.

The dual cut is supported only on arcs

$$\boxed{\theta \geq \alpha + \sum_{i \neq j} \beta_{ij} x_{ij}, \qquad \beta_{ij} = \sum_{t} \bar{\lambda}_{ijt}.}$$

Valid globally

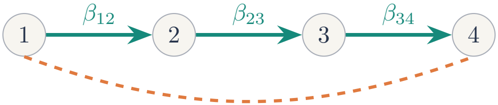
The dual feasible region does not depend on the proposed order.

Tight locally

An optimal dual prices the generating order exactly.

Arc-supported

Every learned coefficient sits on one predecessor arc.



a tool may leave and return across the span

Report: Equation 20 and locality discussion in Chapter 7.

Configurations as columns: PCF, PCF′ and PTF

PCF one configuration per position

$$\min \sum_{t,k} w_t^k \quad \text{s.t.} \quad \sum_C y_C^k = 1, \ \sum_{k\geq 2} w_t^k + \sum_{C\ni t} y_C^1 \geq 1.$$

Without the counting rows the relaxation is $\mathbf{0}$; with them it is exactly $\mathbf{q}$.

PCF′ prefix form

Removes the symmetry of empty positions.

Same relaxation value.

PTF one *transition* per step

$$\min \sum_k \sum_{(C,C')} d(C,C')\, z_{C,C'}^k$$

A column carries a step, so the relaxation cannot avoid paying for it:

$$\text{LP}(\text{PTF}) \ \geq \ q, \quad \text{and can exceed it.}$$

The price

Pricing is NP-hard, and PTF prices coupled *pairs*, so a node costs far more. Rows stay $O(n|T|)$ and fixed.

Report: Section 5.4, Propositions 20–22.

The campaign tests 12 solver configurations

Compact baselines

SSPMF

CATZ–F4

LSS

Reimplementations used for a controlled comparison, not a literature-wide ranking.

Benders regimes

BBC–LP	BBC–LP+T	BBC–K
BBC–LP+F	BBC–K+F	BBC–LP+F+H
BBC–LP+F+C	BBC–LP+F+P	BBC–LP+ACC

1,410 canonical benchmark instances

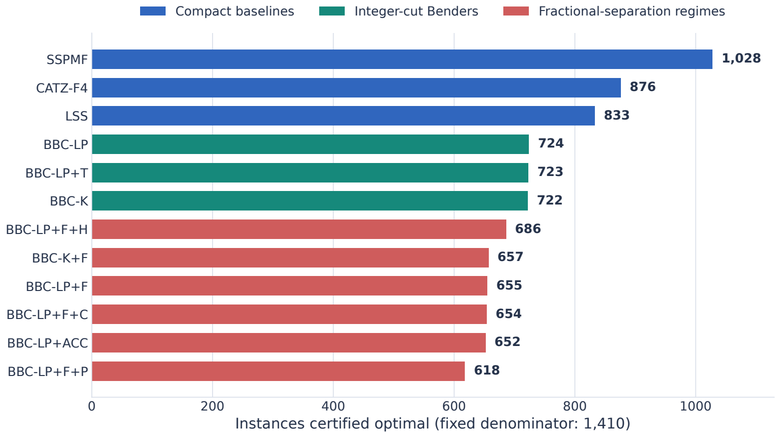
12 solver configurations

16,920 canonical outcomes

1 h per run, fixed denominator

Report: experimental design and Table 5.

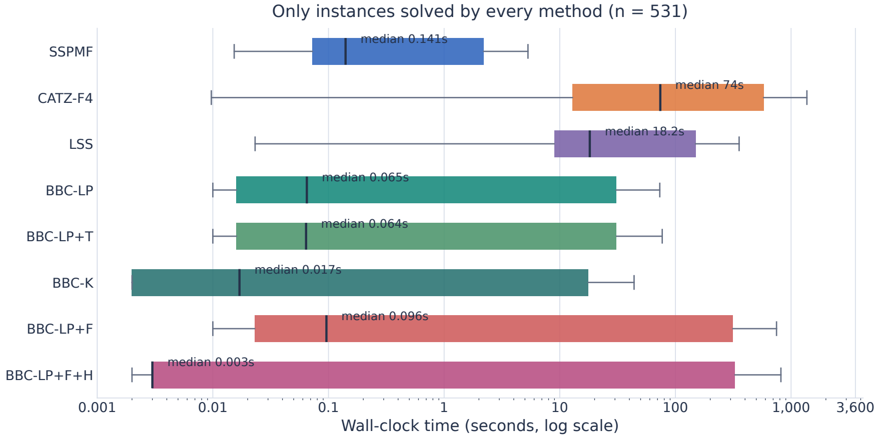
Every compact baseline certifies more than Benders



Each bar uses the same $1{,}410$-instance denominator.

Canonical 16,920-row campaign; report Table 6.

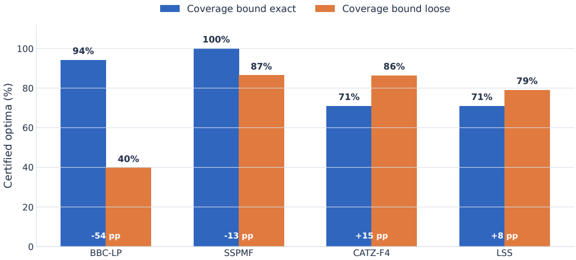
Benders is quick, with a long tail



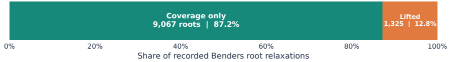
The same 531 instances appear in every box; the horizontal axis is logarithmic.

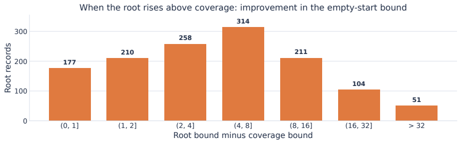
Canonical campaign; common-solved comparison in report Table 7.

Closure drops when the bound misses the optimum



$1{,}107$ instances with a known optimum.





$223/383$ time-limited, known-optimum BBC–LP incumbents already equal the optimum.

Report: Table 10, Observation 13 and the incumbent audit.

Next: pricing tool residency across a span

What the theory isolates

Minimum grouping can force a tool out and back:

$$\text{cost} = q + \text{reinsertions.}$$

What the campaign isolates

The tested master often finds the right order but cannot certify those returns.

Next exact tests

- disaggregate θ by position;
- encode job-union structure over a window;
- test the stronger bound under a matched protocol.

Use the grouping result

- search $K^*, \ldots, K^* + \delta$ groupings;
- flatten each with KTNS; use the orders as seeds.

The work narrows the next experiment; it claims no new SOTA solver.

Report: Discussion and further work, Chapters 7–8.

Selected references

Problem and structure

Tang & Denardo (1988). *Manag. Sci.* 34(7). KTNS optimality.

Crama, Kolen, Oerlemans & Spieksma (1994). *IJFMS* 6. NP-hardness for $b \geq 2$.

Crama & van de Klundert (1999). *Naval Res. Logist.* 46(5). b-approximation of the two-phase method.

Burger, Jacobs, van Vuuren & Visagie (2015). *J. Sched.* 18(2). Industrial study; the open trade-off question.

Exact methods

Laporte, Salazar-González & Semet (2004). *IIE Trans.* 36. Tool-state model (LSS).

Catanzaro, Gouveia & Labbé (2015). *EJOR* 244(3). Formulations F1–F5.

da Silva, Chaves & Yanasse (2021). *IJPR* 59(12). Multicommodity model; LP $= |T| - b$.

Hooker & Ottosson (2003); Codato & Fischetti (2006). Logic-based and combinatorial Benders.

Full bibliography: report, References.

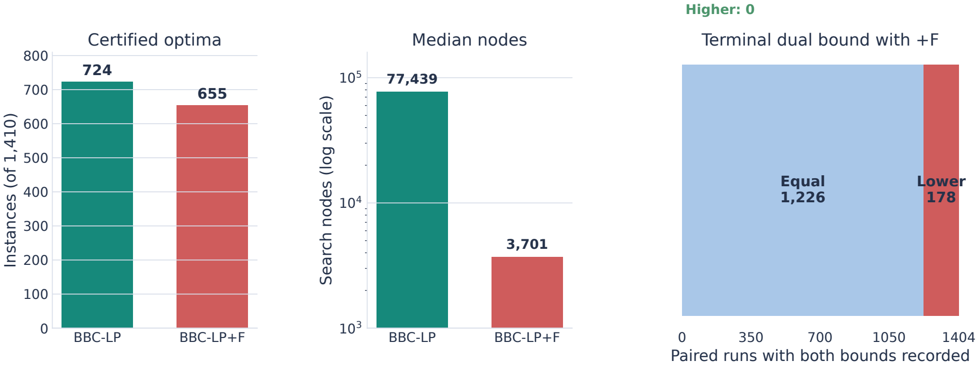
Backup: certified optima by method and family

Configuration	Cat.	Crama	Lap.3	Lap.4	Lap.5	Lap.7	All
SSPMF	62	68	340	330	162	66	1028
CATZ–F4	44	44	340	330	74	44	876
LSS	41	42	340	330	39	41	833
BBC–LP+T	34	34	340	192	84	39	723
BBC–LP	34	35	340	192	84	39	724
BBC–K	35	35	340	190	83	39	722
BBC–LP+F+H	39	37	316	157	97	40	686
BBC–LP+F	33	32	313	156	85	36	655
BBC–LP+F+C	33	32	311	157	85	36	654
BBC–LP+ACC	39	37	286	154	97	39	652
BBC–K+F	33	31	316	156	86	35	657
BBC–LP+F+P	31	32	282	155	83	35	618

Family denominators: $160, 160, 340, 330, 340, 80$; total $1{,}410$.

Report: Table 6.

Backup: fractional cuts spend more, certify less



BBC-LP+F generated 67.0 million fractional cuts. The +F regime also disables presolve.

Against BBC–LP+F: $+H$ certifies 31 more; $+C$ certifies 1 fewer; $+P$ certifies 37 fewer; all three together certify 3 fewer.

Report: Tables 8–9.

Backup: proof and validation ledger

Connected-family proof

$K^* = 3g$	unique feasible pair in each copy
$Z^* = 8g - 5$	coverage bound plus a schedule with no return
$H = 9g - 5$	one forced return per copy plus a matching order
connected	universal tool u lies in every job

Independent checks

$349{,}872$	loading checks against an exact dynamic program
988	instances proved optimal by two methods
$41{,}514$	pairwise method comparisons, none disagreeing
$16{,}920$	canonical campaign outcomes audited

Formal scope and remaining conjectures: `THEOREM_AUDIT.md`.

Report: Appendices A–C and the independent verification program.